# Spam, Speed or Style? Four Ways to Train a Super Smash Bros. Melee Bot

Alejandro García, Teammate Name

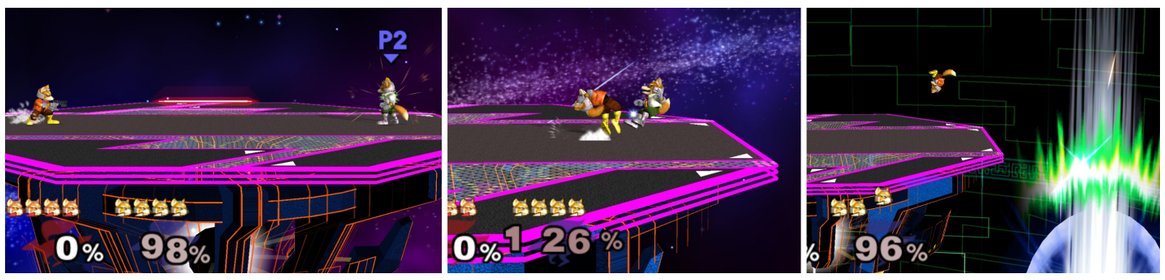
Chalmers University of Technology
Electrical Engineering Department

CHALMERS UNIVERSITY OF TECHNOLOGY

## Motivation

### Which training gives a Fox that is strong *and* human-like?

1 **pure RL**: lasers from across the stage (98%), a rush in (126%), a knockout




4 **imitation + RL**: up-throw, up-air, jab, up-smash, like a person

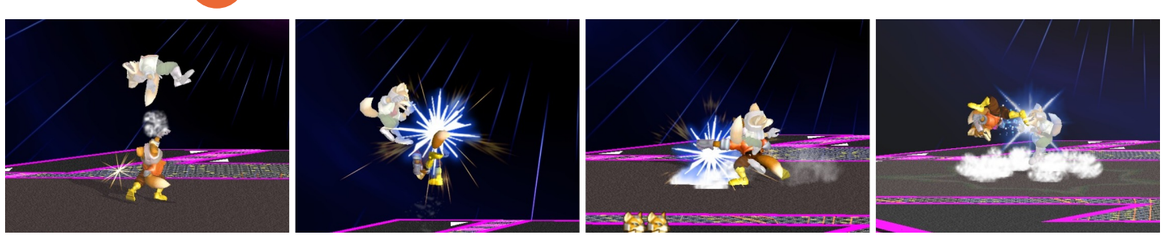


Figure 1: Against the level 9 CPU, captured from the emulator. Pure RL wins every game, but not the way people play.

- **Melee**: 60 frames/s, frame-precise technique. Deep RL has beaten pros [6], but by reacting faster than any human; our first PPO bots won by **exploiting** the game (spamming smash attacks, lasers, shields).
- We train one character, Fox, four ways and compare them on the same stage and rules.

## Hypothesis

**1 Pure RL**
PPO from scratch, sees the game instantly
+ strongest against scripted opponents
– superhuman reactions, exploits the reward

**2 RL + reaction delay**
the same, but sees the game 200 ms late
+ fair: a human reaction time
– much harder to learn, weaker

**3 Imitation**
copies 111k games of strong players
+ human technique: wavedash, L-cancel, shine
– copies mistakes, never tries to win

**4 Imitation + RL**
3, then RL on a leash to the human policy
+ human style and wins
– depends on the human data

**H:** with a human reaction time, 4 beats 2, and pure RL 1 only avoids spam with heavy reward shaping.

## Problem formulation

- **Observe** both players' state from memory (no pixels): position, speed, damage, stocks, action state, character.
- **Act**: 1 2 31 stick + button macros, one every 2 frames; 3 4 a full controller every frame (≈ 69k choices).
- **React**: how late each agent sees the game [5]:




## The four agents




1 2 3 4

| | Policy network | Learns from |
|---|---|---|
| 1 | MLP actor + critic, 2 × 256 ReLU, 0.6 M parameters | CPU curriculum level 1 → 9 [2], then a league of frozen past agents [14] |
| 2 | the same, fed the state of 12 frames ago | starts from 1, then the same league |
| 3 | 3× (LSTM [7] + MLP), width 512, 9.9 M parameters [4] | 111k ranked games of master / diamond Fox players [3] |
| 4 | the same | starts from 3, then self-play, a Ganondorf bot and the level 9 CPU |

## Method: reinforcement learning 1 2

PPO [13, 11] with GAE [12] ($\gamma = 0.995$, $\lambda = 0.95$, $\epsilon = 0.2$), 8 headless emulators in parallel [1, 10]:

$$\mathcal{L}(\theta) = -\,\mathbb{E}_t\Big[\min\left(\rho_t \hat{A}_t,\ \mathrm{clip}(\rho_t, 1 \pm \epsilon)\,\hat{A}_t\right)\Big] \qquad \rho_t = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\text{old}}(a_t \mid o_t)}$$

**Shaped reward** [9], per frame: knockouts first, combos pay, spam costs:

$$r_t = 2\left(\Delta\mathrm{KO}_{\text{opp}} - \Delta\mathrm{KO}_{\text{own}}\right) + 0.003\left(\Delta\%_{\text{opp}} - \Delta\%_{\text{own}}\right) + r_t^{\text{combo}} - r_t^{\text{whiff}} - r_t^{\text{spam}} - r_t^{\text{camp}}$$

- $r^{\text{combo}}$: 0.1 $k$ for the $k$-th hit of a listed combo; $r^{\text{whiff}}$: 0.03 per missed attack, 0.2 per missed smash.
- $r^{\text{spam}}$: the $n$-th repeat of a smash, move or dodge costs $\min(2,\ b\,2^{n-2})$, doubling up to one knockout. 2 acts on $o_{t-12}$.

## Method: learning from humans 3 4

**Behaviour cloning** 3: predict the human's controller 18 frames ahead [4]:

$$\mathcal{L}_{\text{BC}}(\theta) = -\sum_t \sum_{k=1}^{4} \log \pi_\theta\left(a_{t+18}^{(k)} \,\middle|\, h_t,\ a_{t+18}^{(<k)}\right)$$

**RL on a leash** 4: the PPO loss above plus a KL penalty to the frozen imitation policy [8, 14], $\mathcal{L}(\theta) = \mathcal{L}_{\text{PPO}}(\theta) + \beta\,\mathbb{E}_t\big[\mathrm{KL}\big(\pi_\theta \,\|\, \pi_{\text{BC}}\big)(h_t)\big]$, $\beta = 0.5$, $\gamma = 0.997$; reward: stocks and damage.

## Training (preliminary)

| ≈ 91 h | +12 h | 18 h | +3 h |
|---|---|---|---|
| pure RL | RL + delay | imitation | imitation + RL |
| 90 M steps, 8 games | 12 M steps, from 1 | 1.24 B frames, 1 GPU | 10 games, from 3 |

**Training curves of the four agents** (final runs still going)
win rate against the level 9 CPU, stock lead and share of attacks that miss, against training time

- So far: 1 wins 100% against the level 9 CPU; the spam penalties cut smash-attack spam from ∼24 to ∼4 per game. With the 200 ms delay, learning from scratch stalled at CPU level 4–5, so 2 starts from 1.

## Results: final tournament

| | 1 | 2 | 3 | 4 | CPU | Gan. |
|---|---|---|---|---|---|---|
| 1 | – | ? | ? | ? | ? | ? |
| 2 | ? | – | ? | ? | ? | ? |
| 3 | ? | ? | – | ? | ? | ? |
| 4 | ? | ? | ? | – | ? | ? |

Win rate of the row agent. Final Destination, 4 stocks, ? games per pair; CPU: level 9 Fox; Gan.: the Ganondorf PPO bot.

**Tournament figure**
win rates and play style (attacks, combos, spam per minute) of the four agents

## Conclusion

- *(after the tournament)* which recipe wins with a human reaction time, at what cost in style. Next: more stages, games against people.

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youremailhere@student.chalmers.se